A Comprehensive Overview of Audio Language Models

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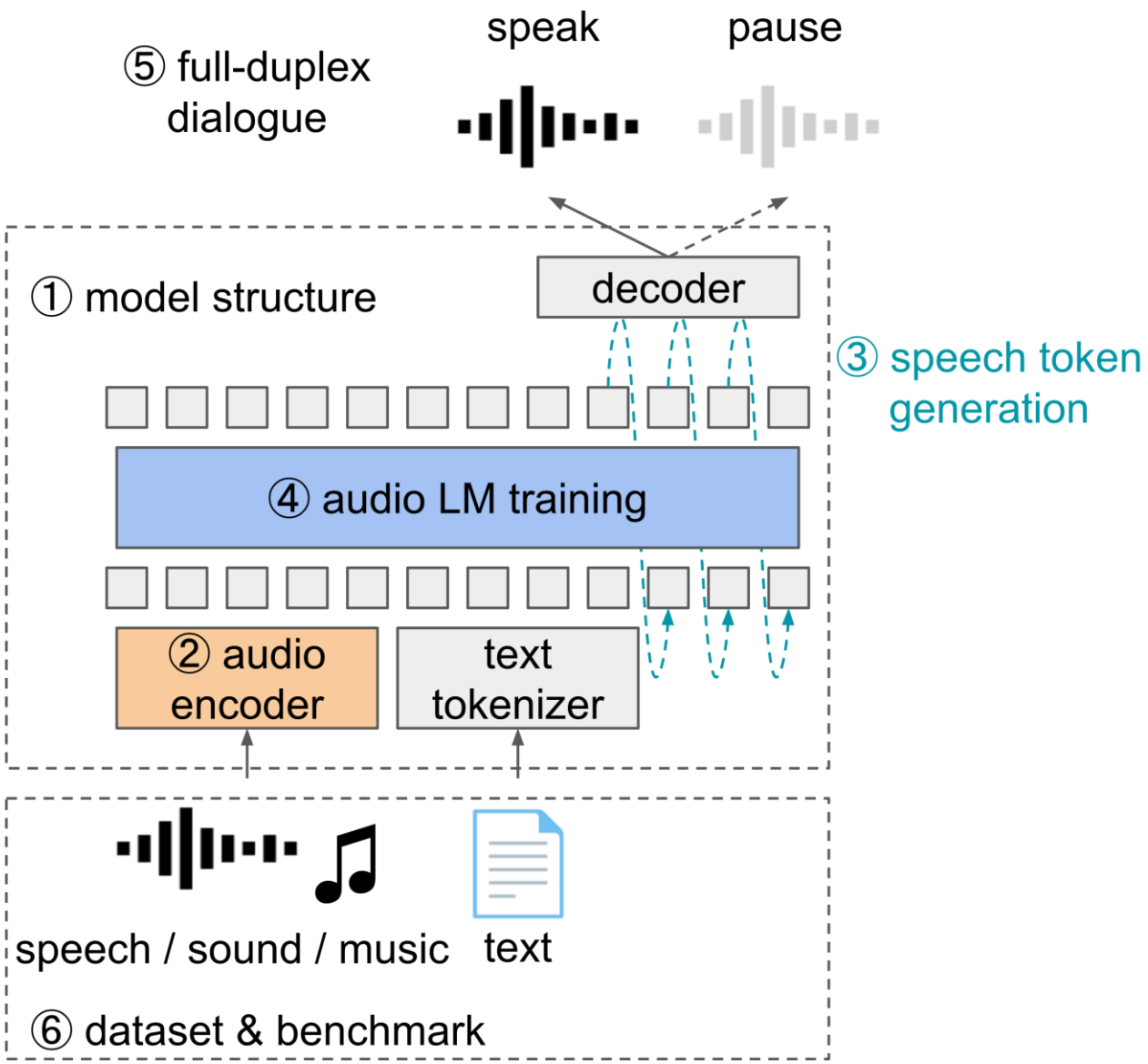
Audio language models

- Audio language models (LMs) process any audio (speech, environmental sound, and music)

- Conventional cascade systems
  - Combine ASR→NLU→TTS
  - Work well for linguistic tasks
  - Limitation in handling para-linguistic information as well as turn-taking cues that cannot be fully represented in text

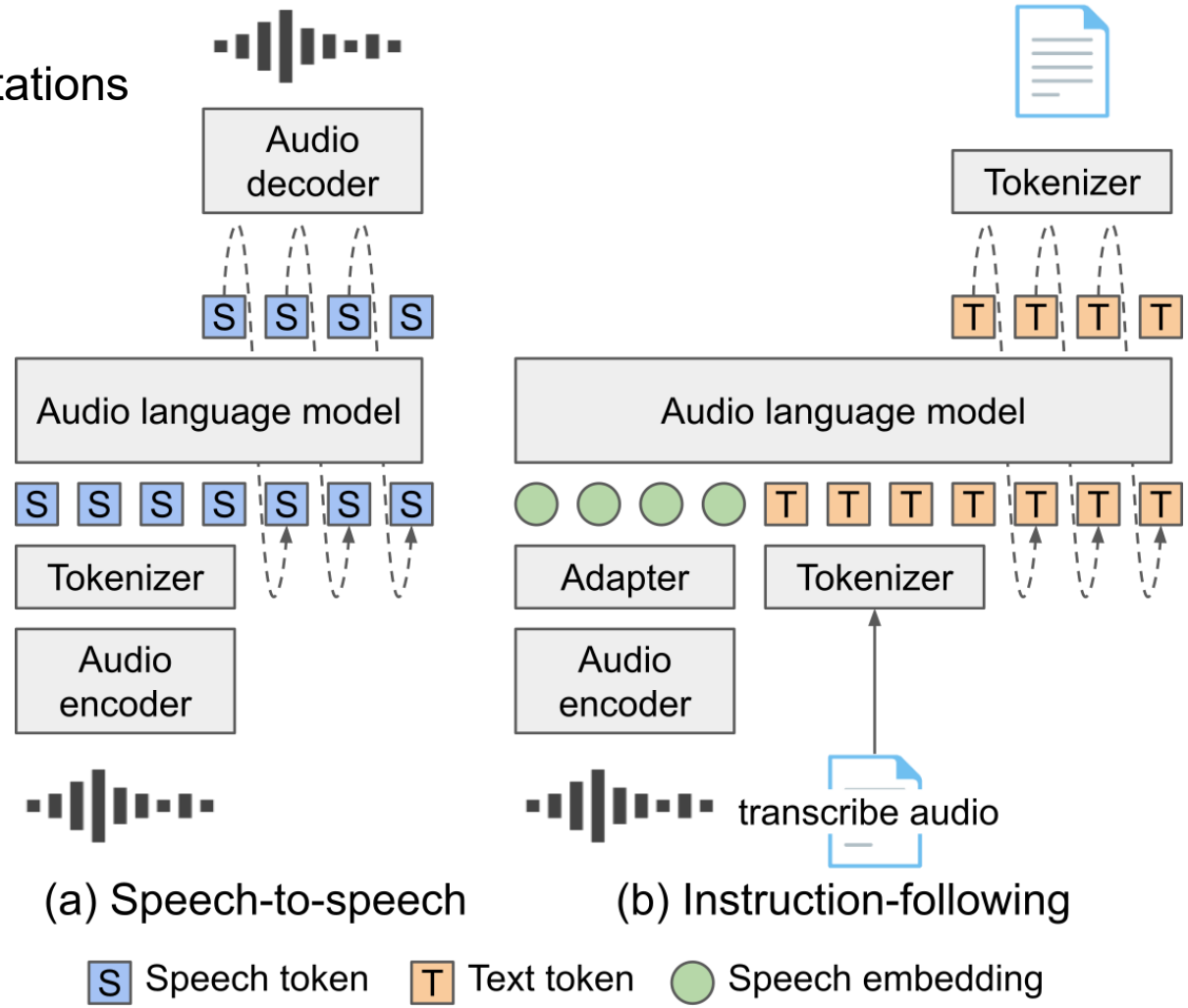
- Recent end-to-end models
  - Understand and generate audio by directly processing audio representations



Model structure

Model structure

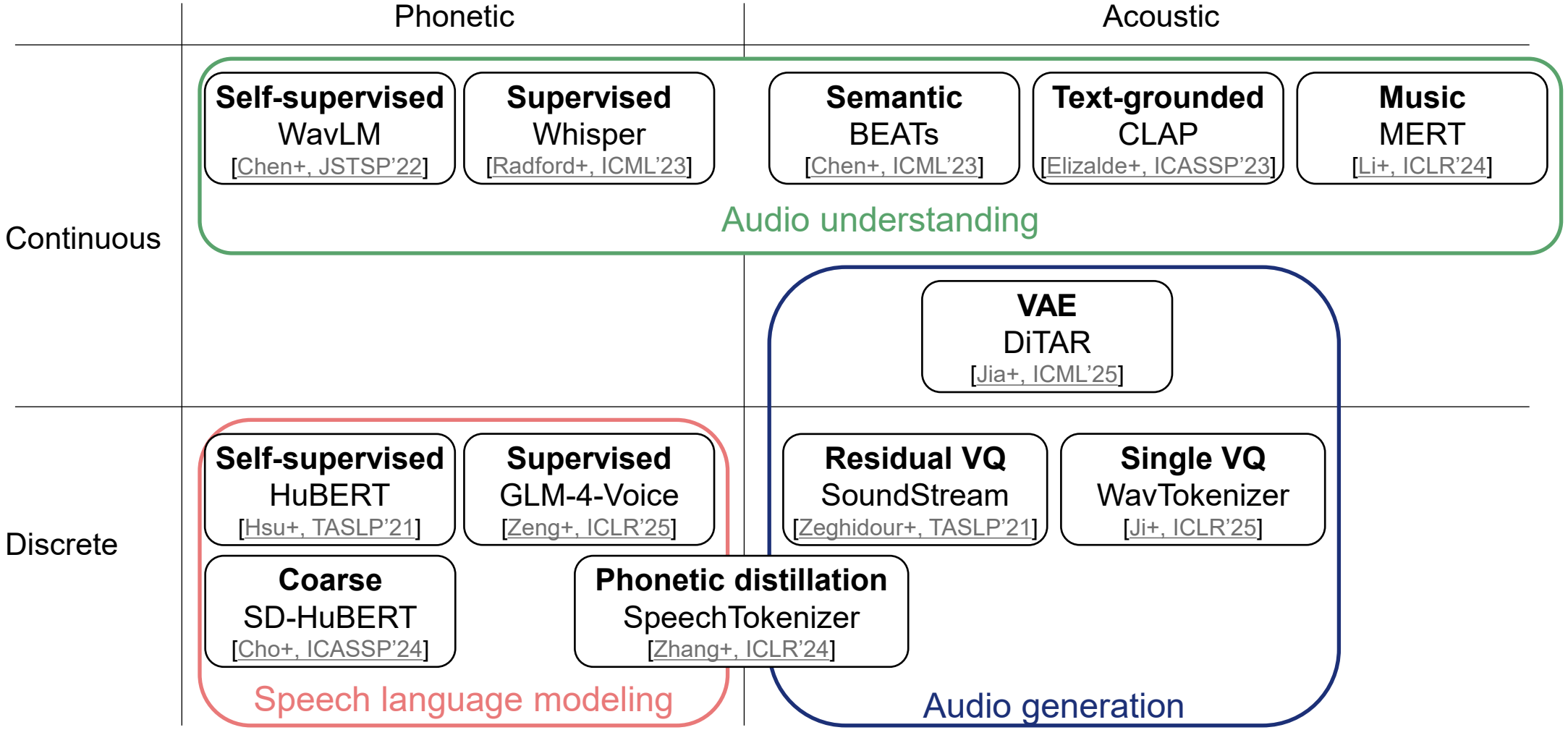
- Speech-to-speech dialogue models
  - Generate continuous or discrete speech representations
  - Then decode them back into speech waveform
  - Enables **spoken conversation**
  - Example: Moshi [Défossez+, arXiv'24]
- Instruction-following audio-to-text models
  - **Output text** responses based on input audio representations, projected by an adapter
  - Designed for reasoning over audio input
  - Example: Qwen2-Audio [Chu+, arXiv'24]



Audio representations

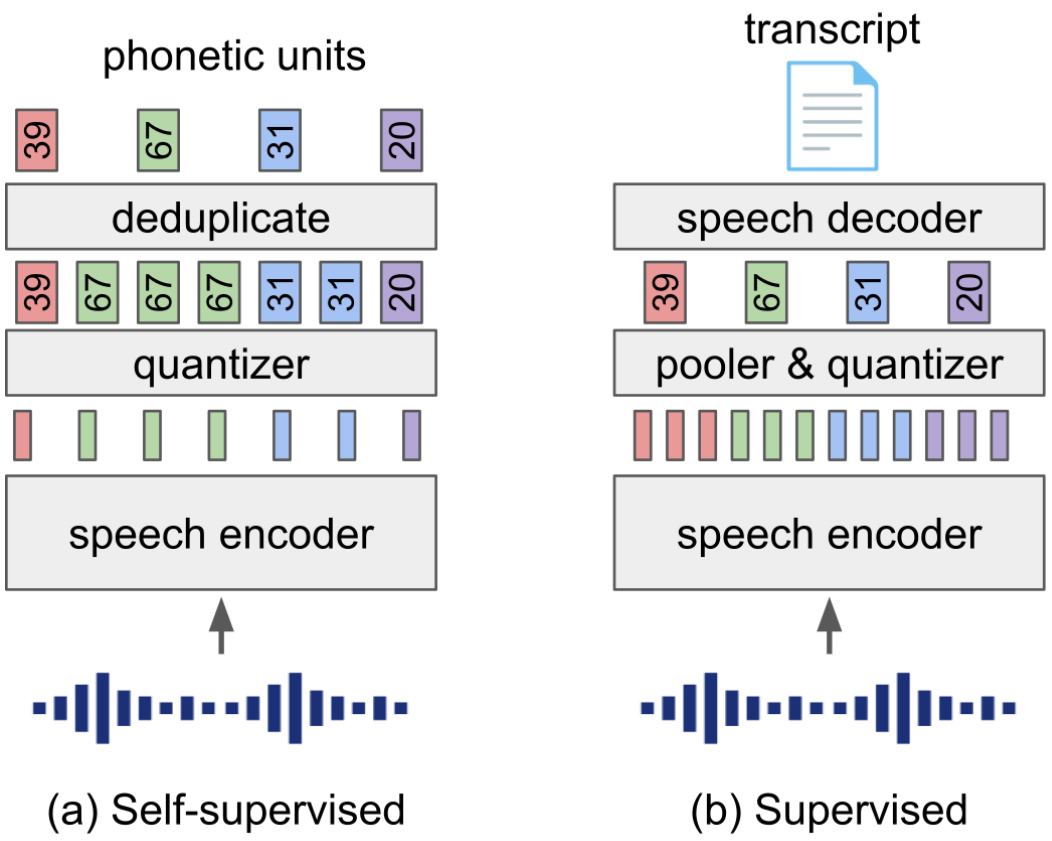
Audio representations

- Can be categorized into four types: **continuous/discrete** × **phonetic/acoustic**



Phonetic token (phonetic units)

- Phonetic token: Clustering indices of speech frame features
  - **Self-supervised**: HuBERT [Hsu+, TASLP'21]
    - Offline K-means clustering
  - **Supervised**: GLM-4-Voice [Zeng+, ICLR'25]
    - Online vector quantization via ASR



(a) Self-supervised (b) Supervised

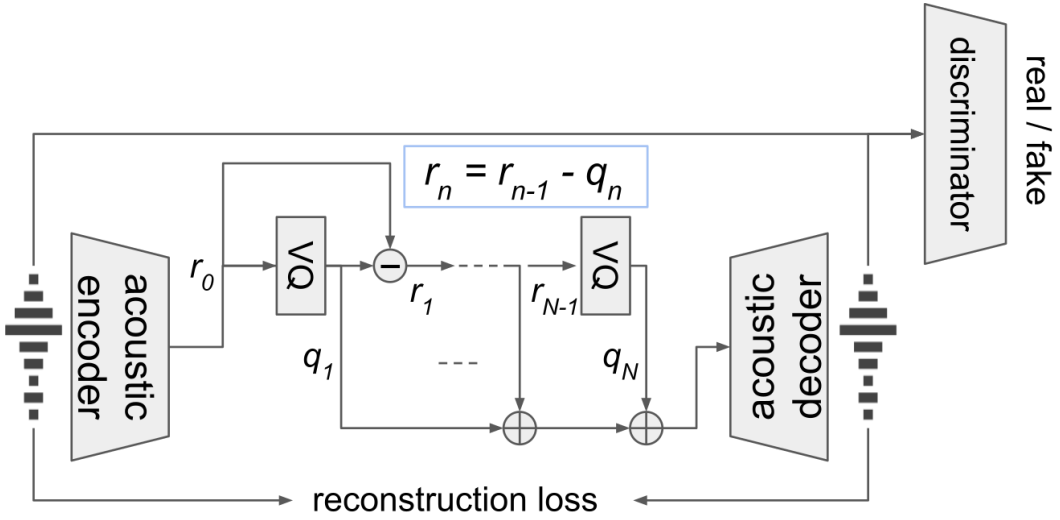
- Encoded information depends on the number of clusters
  - Small codebooks
    - Phonetic tokens align closely with linguistic units
  - Large codebooks
    - Preserve **speaker** [Kharitonov+, NAACL'22] and **prosodic** [Onda+, ASRU'25] information
    - Linguistic understanding performance tends to degrade [Kando+, Interspeech'25]

Acoustic tokenizer (neural audio codec)

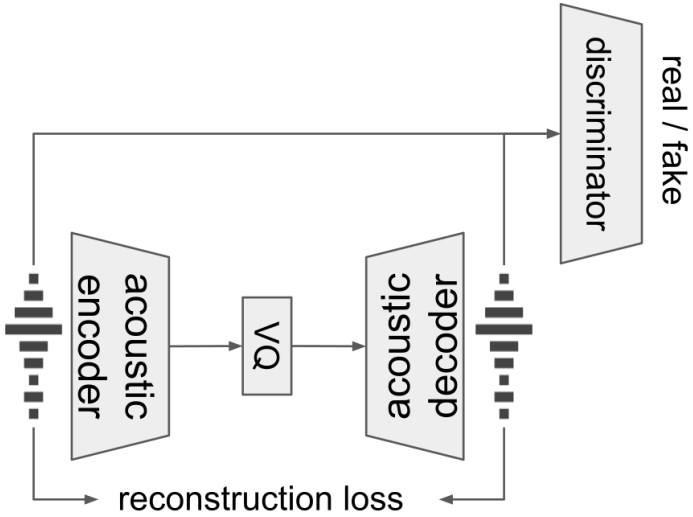
- Autoencoder with intermediate vector quantizers (VQs)
  - **Residual VQ**: SoundStream [Zeghidour+, TASLP'21]
  - **Single VQ**: WavTokenizer [Ji+, ICLR'25]

- Challenge: Codebook collapse in high-dimensional space
  - Only a subset of codes is activated
  - Lowers coding efficiency and reconstruction quality

- Solutions
  - K-means-based codebook initialization [Zeghidour+, TASLP'21]
  - Randomly re-initialize non-activated codes [Dhariwal+, arXiv'20]
  - Code lookup in low-dimensional space [Chiu+, ICML'22]
  - Finite scalar quantization (FSQ) [Mentzer+, ICLR'24]



(a) Residual VQ

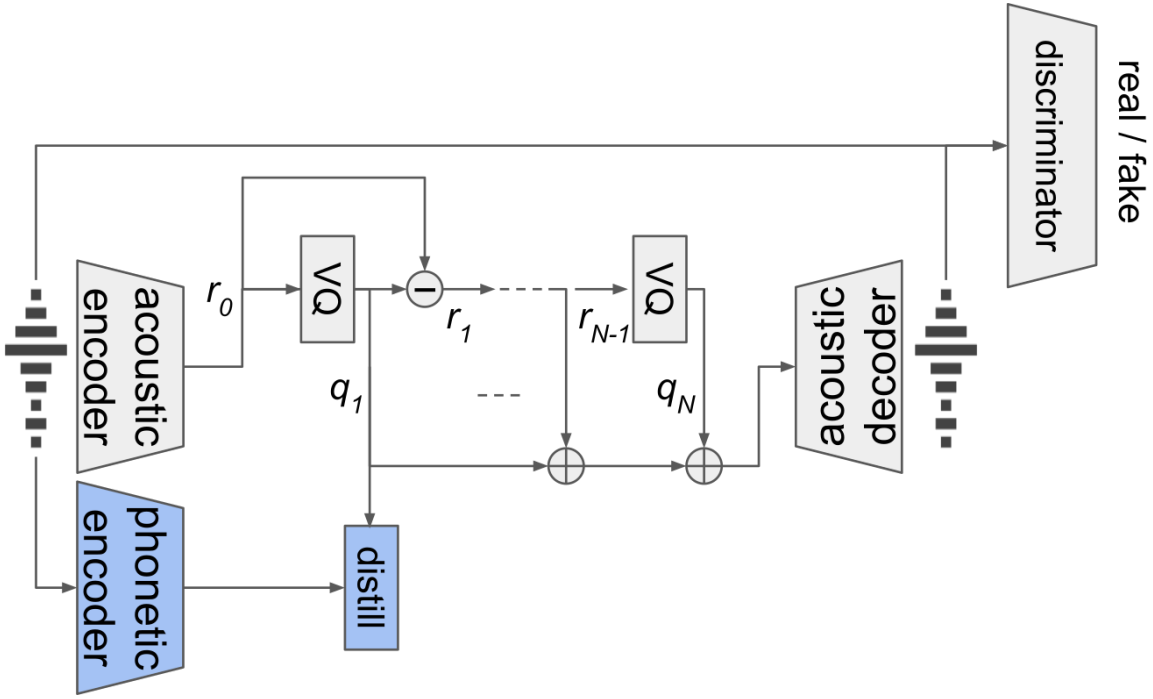


(b) Single VQ

Trade-off between phonetic & acoustic tokens

- Phonetic and acoustic tokens have complementary strengths
  - Acoustic token
    - High-fidelity synthesis
    - Autoregressive generation prone to babbling due to weak correlation with linguistic units
  - Phonetic token
    - Generate consistent linguistic content, as they strongly align with linguistic units

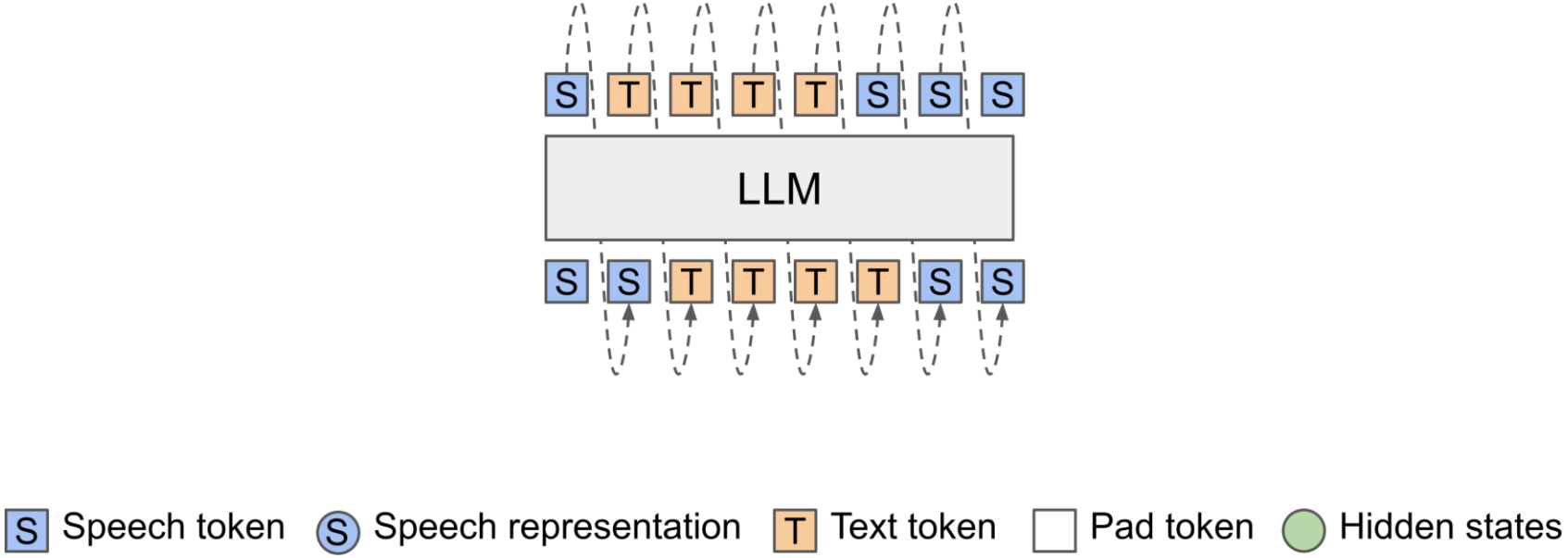
- **Phonetic distillation**: SpeechTokenizer [Zhang+, ICLR'24]
  - Integrates phonetic tokens into residual VQs
  - The first layer distills phonetic content
  - Subsequent layers extract acoustic information



Speech token generation

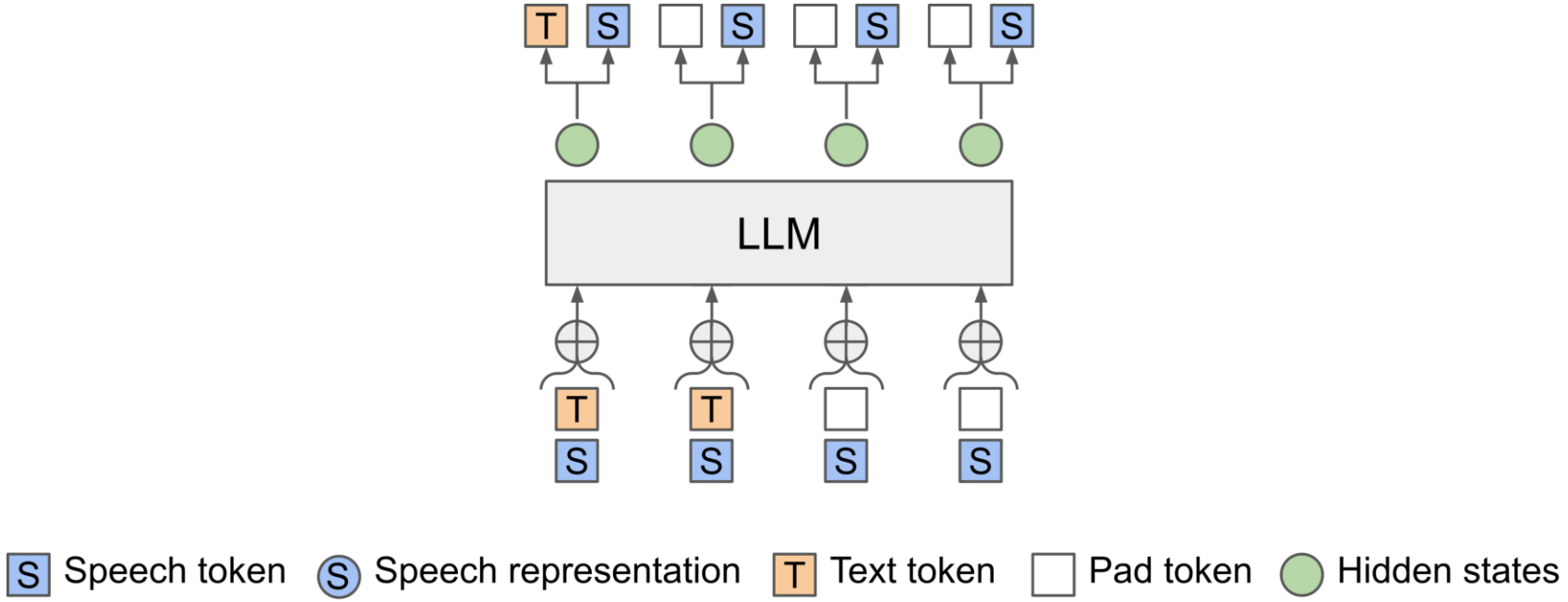
Interleaved generation: GLM-4-Voice [Zeng+, ICLR'25]

- Expands the textual vocabulary with phonetic tokens
- **Alternately generates** speech and text tokens in unified architecture
  - Speech input → transcript → text answer → speech answer
- Text-first thinking improves answer quality by leveraging LLM's reasoning capability
- Intermediate text generation causes high latency
  - Chunk-based streaming: STITCH [Chiang+, arXiv'25]



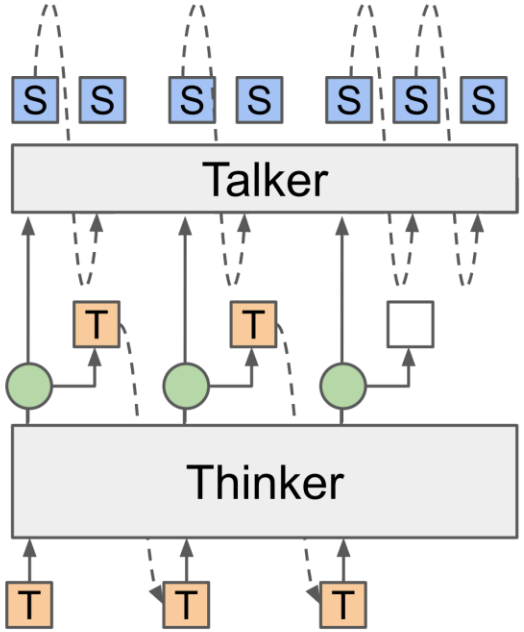
Parallel generation: Moshi [Défossez+, arXiv'24]

- Predicts speech and text tokens using separated modality heads
- Parallel decoding ensures low latency
- Text-prediction guidance (**inner monologue**) improves response quality
- Still underperform interleaved generation due to cross-modal interference in inputs [Wu+, ASRU'25]



Thinker-Talker: LLaMA-Omni 2 [Fang+, ACL'25]

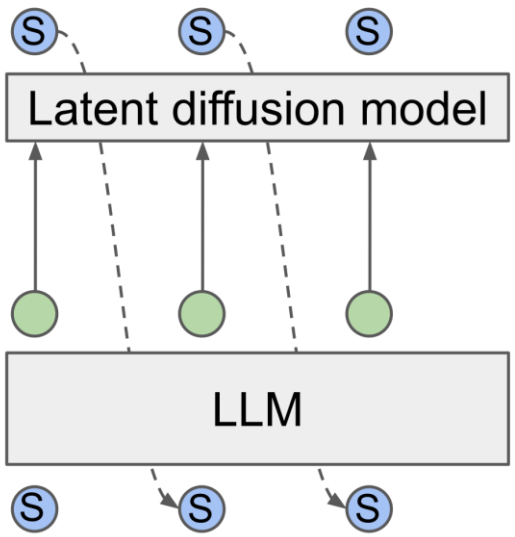
- Separates text reasoning (Thinker) and speech generation (Talker)
- Preserves Thinker's reasoning ability by maintaining text-only I/O
- Talker generates speech tokens from Thinker's hidden states
- Discrete token bottleneck could limit speech quality



S Speech token S Speech representation T Text token Pad token Hidden states

Masked autoregressive: VibeVoice [Peng+, arXiv'25]

- Combines autoregressive LM with **latent diffusion**-based speech generation
- LM sequentially predicts hidden states as diffusion conditions
- Latent diffusion model generates continuous speech representations
- VAE decoder synthesizes waveform from speech representations



S Speech representation T Text token Pad token Hidden states

Training paradigm

Three-stage curriculum learning

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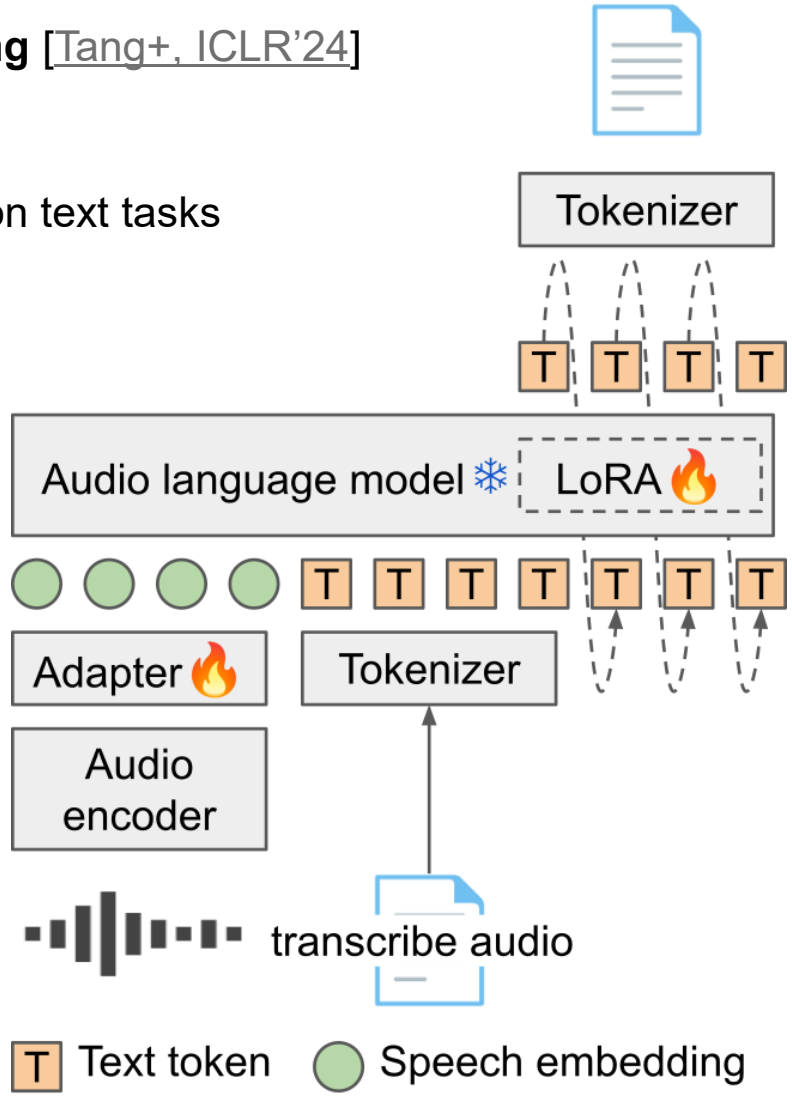
- Detailed training methods differ between instruction-following and speech-to-speech models

1. Text-only pre-training	• Acquires reasoning capability from massive curated text corpus
2. Audio pre-training	• Aligns audio and text in the embedding space • Transfers linguistic knowledge into audio
3. Audio post-training	• Finetunes models for downstream tasks

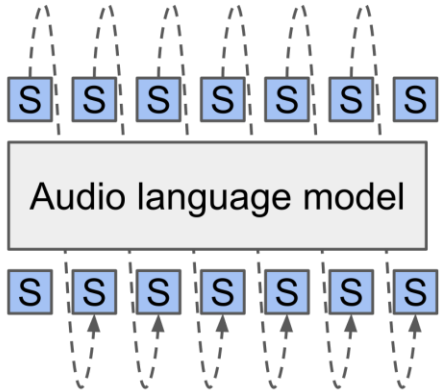
Training for instruction-following models

- ASR-based pre-training
  - Freezes LLM and update only the adapter and LoRA to **preserve text reasoning** [Tang+, ICLR'24]
  - LLMs generalize to unseen audio tasks even when frozen [Wang+, ASRU'23]
  - LoRA improves audio-task performance but may cause catastrophic forgetting on text tasks
  - **Mixing text-only data** mitigates forgetting [Peng+, NAACL'25]
  - Captures only linguistic content
- Audio captioning pre-training: DeSTA [Lu+, arXiv'25], AF3 [Goel+, arXiv'25]
  - Enhance non-linguistic / sound / music understanding
  - Generate captions from transcripts and speaker / emotion metadata
  - Pre-train audio encoder or LLMs to predict captions from audio
- Instruction tuning-based post-training
  - Finetune the model to follow user task instruction via multi-task learning

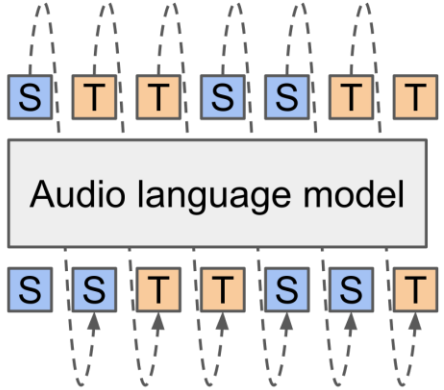


Training for speech-to-speech dialogue models

- Pre-training: Learning speech generation via next-token prediction
  - Speech-only: GSLM [Lakhotia+, TACL'21]
    - Requires no labels
    - Cannot leverage LLM's linguistic knowledge
    - Linguistic performance scales $10^3 \times$ slower than text LLMs [Cuervo+, EMNLP'24]
  - Speech-text interleaving: SpiRit-LM [Nguyen+, TACL'25]
    - Alternate speech and text tokens per words, e.g. "The capital of Japan is <6>"
    - Efficient scaling with both data and model size [Maimon+, COLM'25]
- Post-training: Supervised finetuning
  - Lengthy, itemized sentences are unsuitable for spoken dialogue [Cho+, EMNLP'24]
  - Tune models to generate **concise yet informative** responses [Zeng+, ICLR'25]



(a) Speech-only



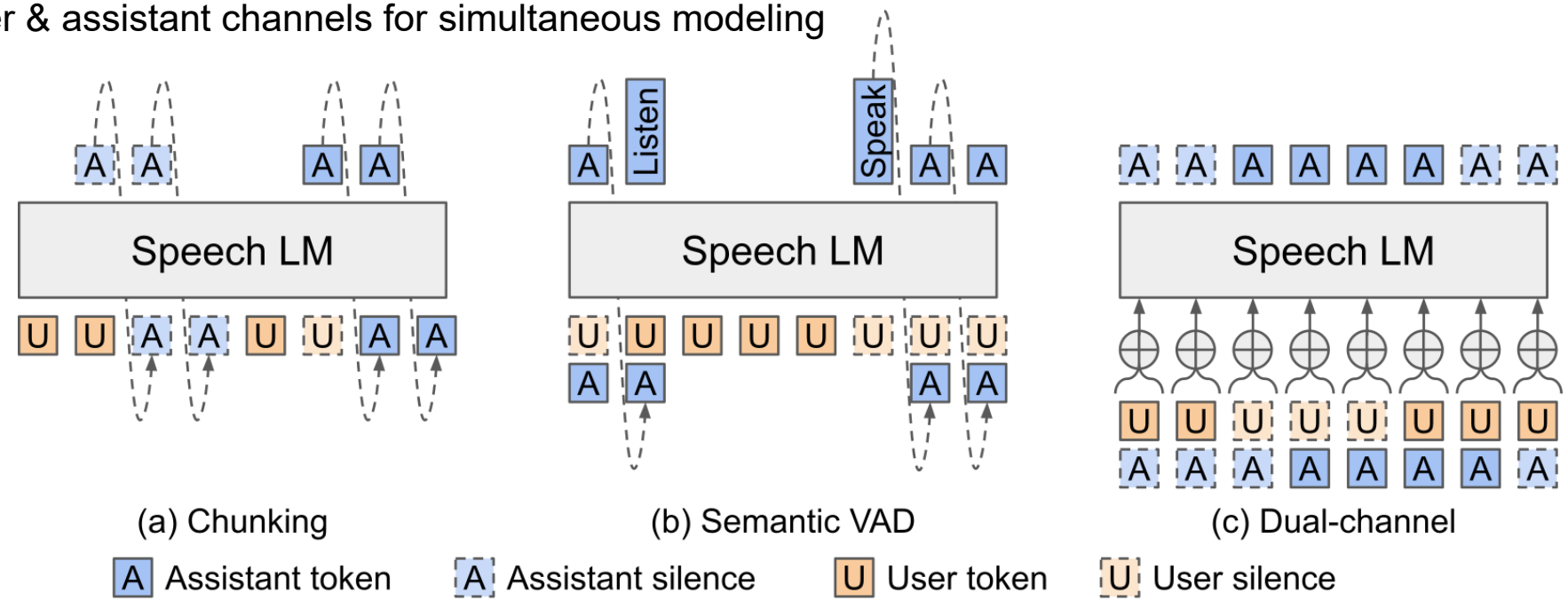
(b) Interleaved

S Speech token T Text token

Full-duplex dialogue

Full-duplex dialogue

- **Chunking**: SyncLLM [Veluri+, EMNLP'24]
  - Alternates user & assistant tokens in fixed-length chunks
  - Generates silence speech tokens to listen
- **Semantic voice activity detection (VAD)**: Freeze-Omni [Wang+, ICML'25]
  - Predicts start-speaking timing from LLM's hidden states
  - Excels at handling speaker's natural pause without interruption [Lin+, ASRU'25]
- **Dual-channel**: Moshi [Défossez+, arXiv'24]
  - Adds user & assistant channels for simultaneous modeling



Datasets and benchmark

Datasets and benchmarks

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- Datasets
  - Synthesize datasets due to limited open dialogue / reasoning datasets
    - Generate question and answer from open audio datasets using text LLM [Rouditchenko+, arXiv'25]
    - Synthesize speech from input text in textual instruction-following dataset
- Benchmarks
  - Evaluate according to the training stage and the target task [Yang+, EMNLP'25]
    - Linguistic knowledge: StoryCloze [Hassid+, NeurIPS'23]
    - Acoustic: Salmon [Maimon+, ICASSP'25]
    - Full-duplex: Full-Duplex-Bench [Lin+, ASRU'25]
    - Reasoning: MMAU [Sakshi+, ICLR'25], Dynamic SUPERB [Huang+, ICLR'25]

Conclusion

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- Overview
  - Audio LMs have rapidly advanced in both understanding and generation
  - They still fall short of human performance on complex reasoning [Sakshi+, ICLR'25]
- Challenges
  - Scaling efficiency
    - Scaling efficiency of audio LMs remains lower than that of text LMs
    - Future direction: Resource-efficient training methods suitable for academic-scale compute
      - Coarse speech token, e.g., Latent speech-text Transformer [Lu+, arXiv'25]
      - Speech-text interleaving
  - Reliability
    - Audio LMs sometimes hallucinate nonexistent sound events (object hallucination) [Kuan+, ICASSP'25]
    - Future direction: Enhance model reliability through improved training algorithms

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